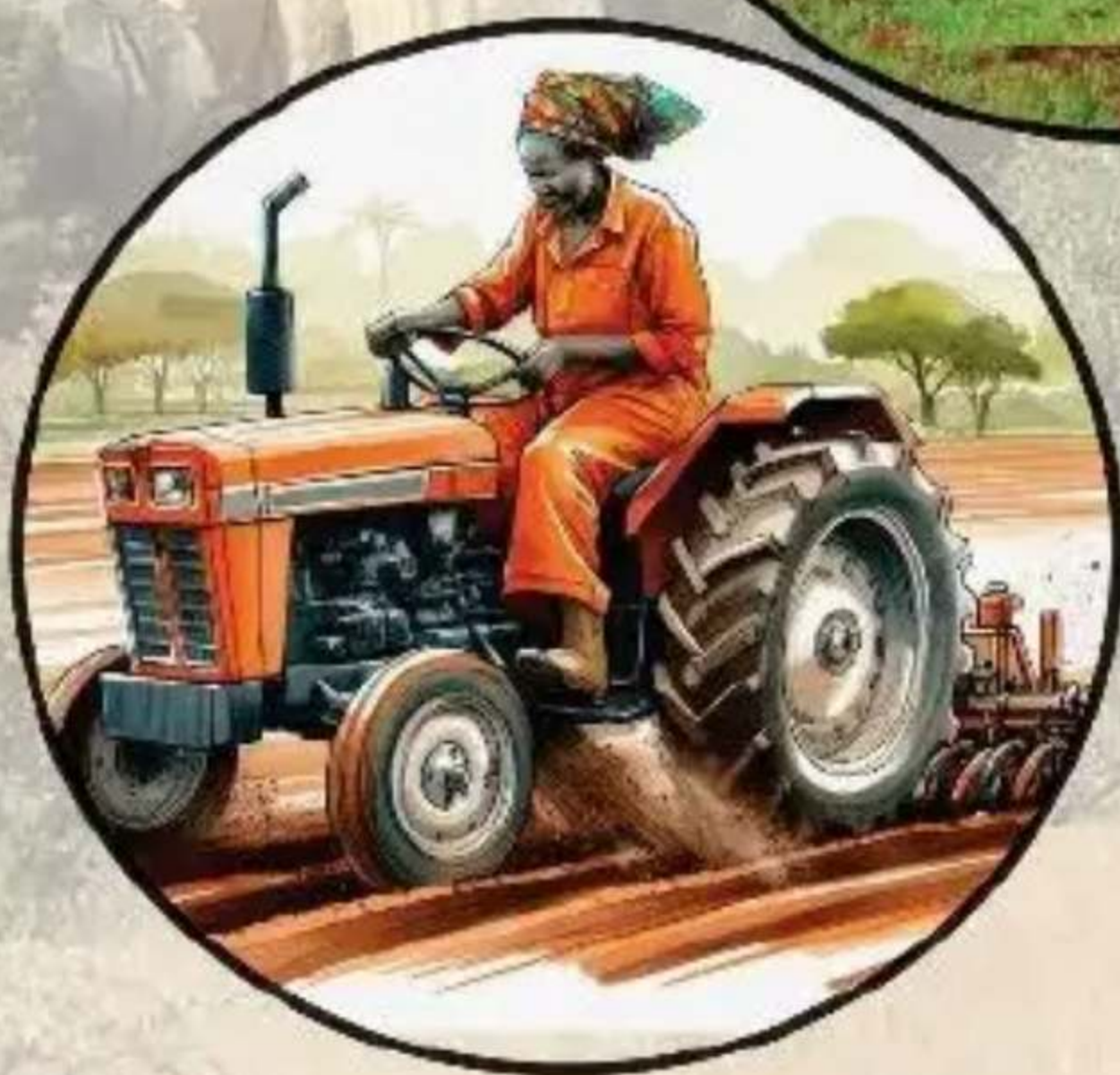




ZIMBABWE

Ministry of Primary and Secondary Education



AGRICULTURE SYLLABUS

2024-2030

FORM 1 - 4

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1.0 PREAMBLE

1.1 INTRODUCTION

The Heritage-based Agriculture is a science learning area that involves theory and practical activities. This four- year learning phase (Form 1 - 4) will provide all learners with opportunities to identify, investigate, solve problems, carry out agricultural activities and assess their viability in a sustainable manner. The learning phase will see all learners being assessed through School Based Continuous Assessment (SBCA) and Summative Assessment (SA).

1.2 RATIONALE

Since Zimbabwe's economy is agro- based, the nation embarked on an agrarian land reform and therefore it is imperative that learners in their diversity acquire necessary agricultural knowledge, skills and positive attitudes. This would enable all learners to be proactive, productive and add value to the national economy. Heritage-based Agriculture enables Learners to value the dignity of labour and harness available resources for enterprise development.

1.3 SUMMARY OF CONTENT

The Form 1 – 4 Heritage-based Agriculture Syllabus will cover theory and practical activities. This four- year learning phase seeks to develop skills in sustainable soil and water management; farm tools and machinery use as well as production of agricultural commodities. The syllabus will help all learners to acquire marketing and value addition skills.

1.4 ASSUMPTIONS

It is assumed that all learners have practical skills and knowledge of:

- growing plants and rearing of animals

- the use of farm tools, implements and machinery
- marketing of agricultural produce
- the use of resources in a sustainable manner

1.5 CROSS-CUTTING THEMES

The following are some of the cross-cutting themes in Agriculture:

- Gender equity
- Disaster risk management
- Health and wellbeing
- Environmental management
- ICT
- Business enterprise skills
- Children's rights and responsibilities
- Climate change

2.0 PRESENTATION OF SYLLABUS

The Heritage-based Agriculture Syllabus is a single document covering Forms 1 - 4. It consists of preamble, aims, objectives, scope and sequence, competence matrix and assessment.

3.0 AIMS

The syllabus aims to help learners to:

- 3.1 appreciate the socio-economic importance of agriculture to the country
- 3.2 develop a positive attitude towards Agriculture and its study as a science
- 3.3 develop positive attitude towards the country's natural resources so as to conserve and use them sustainably
- 3.4 apply psychomotor and communication skills in agriculture
- 3.5 develop leadership and Agri-business skills

- 3.6 develop the ability to solve agricultural problems through the application of indigenous knowledge, scientific skills and new technology
- 3.7 develop innovativeness in agricultural practices including value addition
- 3.8 lay a foundation for advanced studies and a career in the field of Agriculture
- 3.9 contribute to the improvement of nutritional needs and food security for the country

4.0 SYLLABUS OBJECTIVES

Learners should be able to:

- 4.1 describe the importance of agriculture to the national economic development of the country
- 4.2 select suitable techniques, equipment and materials for safe and correct use
- 4.3 relate the environment and climatic conditions to agricultural activities
- 4.4 manage an Agriculture enterprise
- 4.5 design experimental or investigative activities using appropriate techniques
- 4.6 present and interpret information in the form of graphs, diagrams and tables
- 4.7 solve Agricultural problems theoretically and practically
- 4.8 carry out relevant estimations and calculations
- 4.9 design agricultural equipment and structures using local materials
- 4.10 select appropriate techniques to add value to Agricultural produce
- 4.11 demonstrate the ability to conserve natural resources sustainably
- 4.12 apply scientific principles and indigenous knowledge to improve nutritional value and food security
- 4.13 practice conservation techniques to protect the environment
- 4.14 select an agricultural career using the information acquired

5.0 METHODOLOGY AND TIME ALLOCATION

5.1 Methodology

Learner centred and hands on approaches should be used in the development of concepts and skills. These approaches should be inclusive and stimulate curiosity in practical learning. All learners should apply their experiences, knowledge, skills and attitudes independently. Linkage between theory and practice should be prioritised in the teaching and learning of Agriculture.

The following are suggested methods of teaching and learning of Agriculture:

- Project based learning
- Research
- Educational tours
- E-learning
- Experimentation
- Problem solving
- Discovery method
- Demonstrations
- Debate
- Discussions
- Design based learning
- Dramatisation/role play
- Case studies
- Gallery walk
- Resource person(s)
- Simulations
- Song and Dance
- Survey

5.2 Time Allocation

Eight periods of 35 minutes per week should be allocated for adequate coverage of the syllabus. Two double periods for theory and a block of four periods for practicals should be allocated.

NB: Learners should be engaged in at least one Educational Tour and one Seminar per year.

6.0 TOPICS

- 6.1 General Agriculture
- 6.2 Soil and water
- 6.3 Crop husbandry
- 6.4 Animal husbandry
- 6.5 Farm structures and machinery
- 6.6 Agri-business

6.0 SCOPE AND SEQUENCE

6.1 TOPIC 1: GENERAL AGRICULTURE

TOPIC	FORM 1	FORM 2	FORM 3	FORM 4
INTRODUCTION TO AGRICULTURE	• Agriculture of • Branches of Agriculture • Importance of Agriculture • Career opportunities in Agriculture	-----	-----	-----
LAND USE	• Forms of land use • Land tenure • Historical background to land tenure	• Population growth and land use • Farming systems	• Physical farm planning • Crop rotation	-----
ENVIRONMENTAL FACTORS	• Environmental factors	• Modification of adverse environmental factors	• Rainfall: Distribution effectiveness, reliability and intensity	• Natural disasters risk management strategies
NATURAL FARMING REGIONS	• Natural farming of regions of Zimbabwe	-----	-----	-----

FORESTRY	• Forests	• Soft and hard wood • Tree nursery • Tree planting and management	• Timber harvesting and marketing. • Deforestation	• Agro-forestry
WILDLIFE	• Value of wildlife • Wildlife resources • Fauna and Flora • Classification of wildlife	• Sustainable utilisation of wildlife resource • Specially protected plants and animals • Dangerous animals and problem animals	• Indigenous knowledge systems in management of natural resources	• Human and wildlife conflicts

6.2 TOPIC 2: SOIL AND WATER

TOPIC	FORM 1	FORM 2	FORM 3	FORM 4
SOIL FORMATION	• Weathering	-----	• Weathering	-----
SOIL TEXTURE, STRUCTURE AND PROFILE	• Texture and structure	• Soil profile	• Improvement and maintenance of soil structure • Destruction of soil structure	-----
SOIL TYPES	-----	• Composition and properties of each soil type	• Improvement of physical characteristics of soils	-----
SOIL CONSTITUENTS			• Importance of soil components	-----

	-----	-----	• Movement of water • Field capacity • Soil macro and micro-organisms • Importance of living organisms	
SOIL TEMPERATURE	-----	-----	• Influence of soil temperature on plant growth and soil organisms • Modification of soil temperature	-----
SOIL FERTILITY	• Plant nutrients	• Organic and inorganic fertilisers	• Fertiliser application • Soil pH and liming • Soil sampling	• Nitrogen cycle
SOIL EROSION AND CONSERVATION	• Soil erosion	-----	• Conservation methods and structures	-----
WATER LOSS AND SOIL DRAINAGE	• Causes of water loss	-----	• Drainage and water logging • Leaching	-----
WATER CONSERVATION	• Water conservation • Methods of water conservation	-----	• Rain water harvesting and storage	• Water pollution • Water legislation
IRRIGATION	-----	• Importance of irrigation • Sources of water for irrigation	• Methods and types of irrigation • Choice of an irrigation system	• Irrigation equipment

6.3 TOPIC 3: CROP HUSBANDRY

TOPIC	FORM 1	FORM 2	FORM 3	FORM 4
CLASSIFICATION OF PLANTS	Classification of plants	Botanical classification	-----	-----
STRUCTURE OF FLOWERING PLANTS	External structure of a plant	Structure of a flower	Plant anatomy and physiology	-----
PLANT PROCESSES	-----	- Reproduction - Germination	- Water and nutrient uptake - Transpiration - Photosynthesis - Translocation and food storage	- Respiration - Plant tropisms
CROP IMPROVEMENT	-----	-----	Crop breeding	-----
CROP PRODUCTION	- Horticulture - Land preparation - Crop management	- Land preparation - Fruit tree production	- Field crops - Land preparation - Legume and cereal production	Crop management
CROP PROTECTION	- Pests - Diseases - Weeds	- Pests - Diseases - Weeds	- Pests - Disease - Weeds	Agrochemicals

6.4 TOPIC 4: ANIMAL HUSBANDRY

TOPIC	FORM 1	FORM 2	FORM 3	FORM 4
TYPES OF LIVESTOCK	• Types of livestock	• Ruminants and non-ruminants	-----	-----
ANATOMY AND PHYSIOLOGY	-----	• Reproduction in poultry	• Digestive system of a ruminant and non-ruminant	• Reproductive system of a ruminant
NUTRITION	-----	• Livestock nutrients	• Types of feeds	• Maintenance and production rations
SMALL LIVESTOCK PRODUCTION	• Broiler production	• Broiler management • Slaughtering, processing and marketing	• Rearing of rabbits/layers/indigenous chickens	• Slaughtering, processing and marketing
NON-RUMINANTS	-----	-----	• Rearing of non-ruminants	-----
RUMINANTS	-----	-----	-----	• Management of cattle or sheep or goats
ANIMAL HEALTH	• Signs of health and ill-health	• Livestock diseases and Hygiene	• Notifiable livestock diseases	• Animal parasites and immunisation
ANIMAL IMPROVEMENT GENETICS	-----	-----	• Genetics	• Breeding

6.5 TOPIC 5: FARM STRUCTURES AND MACHINERY

TOPIC	FORM 1	FORM 2	FORM 3	FORM 4
FARM IMPLEMENTS	• Implements	• Adjustments of animal drawn implements	• Maintenance	-----
FENCING	• Types of fences	• Fencing materials and tools	• Fencing materials	• Anchors and fencing calculations
FARM BUILDINGS	-----	-----	• Farm buildings • Properties of building materials	• Designing livestock buildings
FARM ROADS	-----	• Siting of farm roads	• Features of farm roads	• Road construction and maintenance
APPROPRIATE TECHNOLOGY	-----	-----	• Irrigation pumps	• Shellers
HARNESSING	• Harnesses: yokes	• Harnesses: breast band and collar	-----	-----

6.6 TOPIC 6: AGRI-BUSINESS

TOPIC	FORM 1	FORM 2	FORM 3	FORM 4
FARM RECORDS AND ACCOUNTS	• Farm records	• Profit and Loss Account	-----	-----
PRINCIPLES OF ECONOMICS	-----	-----	• Opportunity cost and choices • Demand, supply and price	• Diminishing returns • Risk and uncertainty • Decision making
FARM BUDGETING	-----	-----	• Budgets	-----
AGRICULTURAL MARKETING	-----	• Types of markets	• Functions and factors of marketing	• Marketing legislation
AGRICULTURAL COOPERATIVES	• Principles of cooperatives • Types of cooperatives	• Cooperatives	-----	-----

7.0 COMPETENCY MATRIX

FORM 1

7.1 TOPIC 1: GENERAL AGRICULTURE

SUB TOPIC: INTRODUCTION TO AGRICULTURE AND LAND USE

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Introduction to Agriculture	- Define Agriculture - Describe the branches of Agriculture - Discuss the importance of Agriculture - Identify career opportunities in Agriculture	- Agriculture - Branches - Crop Husbandry - Animal Husbandry - Horticulture - Soil Science - Wildlife management - Forestry - Agriculture Engineering - Agribusiness • Importance - social - economic - ecological • Career opportunities	- Defining Agriculture - Visiting various branches of Agriculture - Discussing the social, economic and ecological importance of Agriculture - Identifying career opportunities in Agriculture	- ICT tools - Resource persons

Forms of land use	- describe forms of land use - explain factors limiting land use - identify protected areas in Zimbabwe	- Forms of land use: - Forestry - Wildlife management - Crop and livestock husbandry - Factors limiting land use - Protected areas	- Carrying out community land use survey to determine the main agricultural activities - Discussing the forms of land use in their locality - Explaining factors affecting land use - Identifying protected areas in Zimbabwe	ICT tools
Land tenure	- explain land tenure - describe each land tenure system	- Land tenure: - freehold, - lease hold, - communal and - resettlements	- Discussing land tenure systems - Carrying out surveys on land tenure systems in the community	- ICT tools - Resource person
Historical background to land tenure	- explain the importance of land as a national heritage - explain ownership during precolonial period - discuss the effects of colonial rule on land ownership - justify land reform programme during 3rd Chimurenga/Umvukela - outline resettlement models adopted during	- Pre-colonial, colonial and post-independence land tenure, 3rd Chimurenga/Umvukela land tenure - Resettlement models: A1 and A2	- Discussing the importance of land as a national heritage - Discussing historical land ownership systems in Zimbabwe - Inviting resource persons involved in the 2nd and 3rd Chimurenga/Umvukela to explain the rationale of undertaking land reform	- ICT tools - Resource person

	the agrarian land reform		• Visiting national museums and monuments • Discussing resettlement models adopted during the Agrarian land reform	
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SUB TOPIC: ENVIRONMENTAL FACTORS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Environmental factors	• outline environmental factors influencing agricultural activities • discuss the effects of environmental factors on agricultural activities • explain effects of temperature on water loss • discuss various forms of wilting • explain the causes of wilting • explain effects of temperature on agricultural activities	• Environmental factors: - Wind, light, temperature, rainfall and humidity • Effects of environmental factors on agricultural activities • Loss of water through evaporation • Evapo-transpiration and wilting of crops - Temporary and permanent wilting • Frost damage	• Discussing environmental factors and their effects on agricultural activities • Measuring environmental factors • Visiting weather stations for measurements of environmental factors • Constructing wind breaks, frost barriers	• ICT tools • Weather station

	outline measures that can be taken to minimise effects of adverse temperature	- Frost protection - Shade		
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SUB TOPIC: NATURAL FARMING REGIONS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Natural farming of regions of Zimbabwe	- explain the importance of natural farming regions - describe suitable farming systems for each region	- Natural farming regions: - Natural region 1 - Natural region 2 - Natural region 3 - Natural region 4 - Natural region 5	- Carrying out a survey to determine farming activities within their locality - Discussing the importance of demarcating Zimbabwe into farming regions	- ICT tools - Map templates - Pictures

SUB TOPIC: FORESTRY

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Forests	explain social, economic, cultural and ecological importance of forests	- Forests - Indigenous and exotic trees	Discussing the importance of forests	- ICT tools - Resource person - Pictures

	• identify major forests in Zimbabwe • identify indigenous and exotic timber trees • identify exotic timber trees grown in Zimbabwe		• Identifying indigenous and exotic trees in their locality • Labelling identified indigenous and exotic trees	
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SUB TOPIC: WILDLIFE

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Wildlife	• explain socio-economic, cultural and ecological importance of wildlife • identify flora and fauna within their locality • classify wildlife according to feeding habits • identify wild animals classified as the big five	• Value of wildlife • Fauna and flora • Classification -feeding habits -big five	• Discussing socio-economic, cultural and ecological importance of wildlife • Surveying on wildlife resources in Zimbabwe • Identifying flora and fauna within their locality • Classifying animals according to feeding habits • Educational touring	• Textbooks • ICT tools • Wildlife • Pictures

7.2 TOPIC 2: SOIL AND WATER

SUB TOPIC: SOIL FORMATION

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Weathering	- describe the role of weathering in soil formation - describe the types of rocks from which soil is formed - explain the agents of weathering - discuss factors influencing soil formation	- Weathering - Types of rocks	Identifying rock samples	- ICT tools - Rock samples - Pictures

SUB TOPIC: SOIL TEXTURE, STRUCTURE AND PROFILE

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Texture and structure	- identify soil particles according to increasing order of size - explain the significance of soil texture to crop growth - describe soil structure - distinguish single grain from crumb structure - distinguish between soil structure and soil texture	- Soil texture - Soil structure	- Feeling different soil samples to determine texture - Carrying out sedimentation experiments - Conducting experiments to determine the effects of texture on emergence of seeds - Experimenting on the characteristics of soils	- ICT tools - Soil samples - Pictures - Sedimentation apparatus

SUB TOPIC: SOIL FERTILITY

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Plant nutrients	- explain the functions of major and minor plant nutrients to crop growth - describe effects of nutrients on plant growth	Major and minor nutrients	- Listing major and minor nutrients - Identifying symptoms of nutrient deficiencies and over supply	- Fertilizers - ICT tools - Pictures showing affected plants

SUB TOPIC: SOIL EROSION AND CONSERVATION

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Soil erosion	- identify types of soil erosion - describe the causes and consequences of soil erosion - describe the prevention and control of soil erosion	Soil erosion	- Identifying signs and types of soil erosion - Experimenting the effects of soil erosion on different soil types	- ICT tools - Experiment apparatus - Eroded places - Pictures showing different types of erosion

SUB TOPIC: WATER CONSERVATION

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Water conservation	- explain the importance of water conservation - describe methods of conserving water on arable lands	Water conservation	- Discussing the importance of water conservation on arable land - Implementing water conservation measures on arable land	- ICT tools - Resource person EMA/AGRITEX

7.3 TOPIC 3: CROP HUSBANDRY

SUB TOPIC: CLASSIFICATION OF PLANTS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Classification of plants	- classify plants according to edible parts - classify plants according to life cycle	- edible plant parts: leaf, root, tuber, bulb, fruit, stem, seed - Life cycle: annual, biennial, perennial	- Collecting samples of parts eaten - Identifying crops according to their classes - Collecting samples, pressing and pasting onto a folder according to their classes	- ICT tools - Plant specimens

SUB TOPIC: STRUCTURE OF FLOWERING PLANTS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
External structure of a plant	- identify parts of a flowering plant - state functions of each part of a flowering plant	- external parts of a plant - Functions of plant parts	- collecting maize and bean plants - identifying external plant parts - drawing and labelling external parts of a flowering plant - tabulating parts and functions of a flowering plant	- ICT tools - Plant specimens

SUB TOPIC: CROP PRODUCTION

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Horticulture	- identify branches of horticulture - explain the importance of horticulture	- Branches of horticulture - Importance of horticulture	- Identifying horticultural activities in the local community and relate them to their branches - Discussing the importance of horticulture	- ICT tools - Plant specimens
Land preparation	- state reasons for land preparation - prepare seed beds	Seed bed preparation	- discussing reasons for land preparation - preparing a seed bed	- ICT tools - Land preparation tools
Crop management	establish and manage vegetable crops	- Sowing/planting - Management practices - Marketing NB: One vegetable to be grown in each of the following groups: leaf/root/legume/bulbs/tubers/fruit crops according to recommended spacing	Growing leaf/root/legume/fruit crops according to recommended spacing Managing a vegetable crop up to maturity	ICT tools

SUB TOPIC: CROP PROTECTION

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Pests	- explain the effects of pests on crops - classify pests according to their feeding habits	- Effects of pests on crops - Classification of pests	- Discussing the effects of pests on crops - Observing nature of crop damage in the garden - Scouting in the garden/field to identify pests - Classifying pests according to their feeding habits	- ICT tools - Samples and Specimens
Diseases	- classify plant diseases according to causal organisms - describe the effects of plant diseases on crops.	- Plant diseases - Effects of plant diseases	- Identifying crop damages by diseases - Discussing causes of plant diseases	- s - ICT tools - samples and Specimens
Weeds	- discuss the harmful and beneficial effects of weeds - differentiate annual from perennial weeds	- Effects of weeds - classification of weeds	- Collecting weeds in the local area by classes - Preserving samples of weeds	- ICT tools - Samples and Specimens

7.4 TOPIC 4: ANIMAL HUSBANDRY

SUB TOPIC: TYPES OF LIVESTOCK

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Types of livestock	- Name types of livestock - explain the importance of livestock	- Poultry - Fish - Ruminants - Non ruminants - Importance of animals	- Identifying the types of livestock - Discussing the importance of livestock - Compiling a list of products and by-products of livestock	- ICT tools - Pictures of livestock and livestock products - Livestock

SUB TOPIC: SMALL LIVESTOCK PRODUCTION

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Broiler production	- state the breeds of broilers - explain the importance of rearing broilers - describe the housing requirements for broilers - rear broilers	- Importance of broilers - Breeds of broilers - Housing of broilers - Types of brooders	- Debating on the advantages of keeping broiler chickens - Comparing the characteristics of different breeds - Designing a brooder and deep litter - Rearing broilers	- ICT tools - Pictures - Broilers

SUB TOPIC: ANIMAL HEALTH

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Signs of health and ill-health	- define animal health - distinguish between health and unhealthy farm livestock	- Animal health - Signs of health and ill health	- Observing signs of ill-health in farm animals - Comparing healthy and unhealthy animals	- ICT tools - Pictures - Vet officer - Animals

7.5 TOPIC 5: FARM STRUCTURES AND MACHINERY

SUB TOPIC: FARM IMPLEMENTS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Implements	- list 5 tillage implements - draw and label the parts of a mould board plough - explain functions of parts of a mould board plough	- Mould board plough - Cultivator - Harrow - Planter - Ridger	- Identifying the tillage implements - Drawing and labelling a mould board plough - Describe the functions of a mould board plough - Touring a farm with tillage implements	- ICT tools - Pictures of implements - Tillage implements

SUB TOPIC: FENCING

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Types of fences	- explain the reasons for fencing - identify different types of fences	- Purpose of fencing - Types of fences	- Discussing the reasons for fencing - Identifying different types of fences within the locality	- ICT tools - Pictures of types of fence - Types of fence

SUB TOPIC: HARNESSING

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Harnesses	- list the types of yokes - describe the types of yokes - draw and label parts of yokes - describe characteristics of wood used to make yokes	- Yokes - Characteristics of wood for making yokes	- Describing types and characteristics of yokes - Designing harnesses for cattle - Constructing yoke models	- ICT tools - Pictures of yokes - Yokes

7.6 TOPIC 6: AGRI-BUSINESS

SUB TOPIC: FARM RECORDS AND ACCOUNTS

TOPIC	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Farm records	- describe the importance of farm records - differentiate physical from financial records	Farm records	- Discussing the importance of farm records - Compiling records for Agriculture projects at the school - Educational touring of local farms	- ICT tools - Farm records

SUB TOPIC: AGRICULTURAL COOPERATIVES

TOPIC	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Principles of cooperatives	- explain the principles of Agricultural cooperatives - discuss how cooperatives are formed	Cooperatives	- Discussing the principles of cooperatives - Discussing how cooperatives are formed	- ICT tools - Resource person

Types of cooperatives	explain the types of Agricultural cooperatives in Zimbabwe	types of cooperatives in agriculture	discussing various types of cooperatives in the farming sector	- ICT tools - Resource person from cooperative office
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FORM 2

TOPIC 1: GENERAL AGRICULTURE

SUB TOPIC: LAND USE

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Population growth and land use	explain the effects of population on land use	Land pressure	- Discussing the effects of population on land use - Calculating population density in their local community	- ICT tools - Demographic maps
Farming systems	Describe different farming systems	- Mixed farming - Monoculture - Intercropping	- Discussing farming system - Educational touring of local farms	- ICT tools - Local farms

SUB TOPIC: ENVIRONMENTAL FACTORS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Modification of adverse environmental factors	describe ways of reducing effects of environmental factors on Agricultural activities	ways of reducing adverse environmental factors	- Practising shading, mulching, pot holing, manuring, tie ridging, watering and conservation tillage - Visiting green houses	- ICT tools - Local modifications such as wind breaks, mulches, manure

SUB TOPIC: FORESTRY

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Soft and hard wood	distinguish between soft and hard wood	Soft and hard wood	Identifying soft and hard wood timber species	- ICT tools - Resource person from forestry - Pictures - Nursery site
Tree nursery	- describe factors influencing choice of a nursery site - establish and manage tree seedlings in a nursery	- Factors influencing choice of a nursery site - Tree nursery establishment and management	- Selecting a nursery site for raising seedlings - Raising tree seedlings - Visiting established nurseries	- ICT tools - Resource person from forestry - Pictures - Nursery site
Tree planting and management	Establish a tree plantation	Tree plantation: -establishment	Discussing the establishment and	ICT tools

	Discuss the management practices of a tree plantation	-management	management of tree plantation Demonstrating proper tree planting and management skills	- Resource person - Woodlot plantation
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SUB TOPIC: WILDLIFE

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Sustainable utilisation of wildlife resources	explain sustainable methods of wildlife utilisation	- Wildlife resources - Sustainable utilisation	- Discussing sustainable methods of wildlife utilisation - Watching documentaries	- ICT tools - Protected areas
Specially protected plants and animals	identify special plants and animals in Zimbabwe	endangered plant and animal species	- Listing specially protected plants and animals - Watching documentaries	- ICT tools - Protected areas - pictures of specially protected plant and animal species
Dangerous and problem animals	- identify dangerous and problem animals in Zimbabwe - describe ways of dealing with dangerous and problem animals in Zimbabwe	Dangerous and problem animals	- Surveying on local dangerous and problem animals - Demonstrating ways of dealing with dangerous and problem animals in Zimbabwe - Watching documentaries	- ICT tools - Protected areas - Pictures of dangerous and problem animals

TOPIC 2: SOIL AND WATER

SUB TOPIC: SOIL TEXTURE, STRUCTURE AND PROFILE

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Soil profile	- describe soil profile with the aid of a diagram - describe the appearance and composition of each horizon - discuss the significance of each horizon to crop growth - explain the importance of soil profile	- Soil profile - Soil profiling	- Describing soil profile with the aid of a diagram - Drawing and labelling a soil profile - Digging a profile pit - Identifying the horizons up to the maximum depth of 1.5m	- ICT tools - Road side excavations - Gullies - Open pits - River beds - Valley sides

SUB TOPIC: SOIL TYPES

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Composition and properties	- identify different soil types - explain the composition of each soil type - compare the composition of soil types - compare the properties of different soil types	- Soil types: - Sand soil - Loam soil - Clay soil - Composition and properties	- Collecting different soil samples - Identifying the soil types - Carrying out experiments to verify different properties of soil type	- ICT tools - Soil samples - Experiment apparatus

SUB TOPIC: SOIL FERTILITY

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Organic and inorganic fertilisers	- distinguish between organic and inorganic fertilisers - describe types of organic fertilisers - describe types of inorganic fertilisers - differentiate compound from straight fertilisers	- Organic fertilisers - Inorganic fertilisers	- Examining organic and inorganic fertilizers - Identifying organic and inorganic fertilisers - Identifying straight and compound fertilisers	- ICT tools - Fertilisers - Agronomist

SUB TOPIC: IRRIGATION

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Importance of irrigation	discuss the importance of irrigation	Importance of irrigation	- Discussing the importance of irrigation - Touring an irrigation scheme	- ICT tools - Irrigation equipment and systems
Sources of water for irrigation	- list sources of water suitable for irrigation - determine the suitability of water for irrigation	- Sources of irrigation water - Water quality	- Discussing the different sources of water for irrigation - Testing for impurities in water	- ICT tools - Local water sources - Irrigation water

TOPIC 3: CROP HUSBANDRY**SUB TOPIC: CLASSIFICATION OF PLANTS**

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Botanical classification of plants	- differentiate monocotyledonous from dicotyledonous plants - state botanical classes of crops	- Monocotyledonous and dicotyledonous plants - Botanical classes: - Legumes - Brassica - Solanaceous - Cereals(cereals) - Cucurbits	- Differentiating classes of crops - Categorising crops according to botanical classes - Carrying out field tours	- ICT tools - Flower samples/specimens - Pictures

SUB TOPIC: STRUCTURE OF FLOWERING PLANTS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Structure of a flower	- draw the structure of maize and bean flowers - label the parts of a flower - state functions of each part of a flower	- maize and bean flowers - Structure of a flowers - Functions of flower parts	- Collecting maize and bean flowers - Identifying parts of a flower - Drawing and labelling parts of a flower - Describing functions of flower parts	- ICT tools - Flower samples/specimens - Pictures

SUB TOPIC: PLANT PROCESSES

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Reproduction	- differentiate between sexual and asexual reproduction - state advantages and disadvantages of sexual and asexual reproduction - describe pollination of maize and bean flowers - describe fertilisation process in plants	- Reproduction - Pollination and fertilisation	- Discussing sexual and asexual reproduction - Discussing advantages and disadvantages of sexual and asexual reproduction - Discussing the fertilisation process in plants	- ICT tools - Samples and specimens of reproductive organs

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
	explain different methods of asexual reproduction		- Observing and differentiating maize and bean flowers - Watching video simulations on fertilisation process in plants - Demonstrating methods of asexual reproduction such as budding, layering	
Germination	- state requirements for seed germination - differentiate between emergence and seed germination - identify internal and external parts of a maize and bean seed	Germination	- Conducting experiments to determine the requirements for conditions necessary for seed germination - Drawing diagrams to illustrate germination and emergence - Dissecting maize and bean seeds - Observing the internal parts and external parts of a bean and maize seeds - Identifying the parts of a bean and maize seeds	- ICT tools - Specimens

SUB TOPIC: CROP PRODUCTION

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Orchard Establishment	• Explain the factors which influence site selection • Describe orchard land preparation • Explain the usefulness of different planting patterns • Peg planting stations • Prepare a planting hole • Plant fruit trees • Calculate amounts of fertilizers required for orchard crops • Design an irrigation schedule • Discuss the reasons for pruning	• Site selection • Land preparation • Planting patterns • Orchard pegging • Planting holes Planting • management practices • harvesting • marketing	• Discussing the factors influencing site selection • Preparing land for an orchard • Discussing the usefulness of different planting patterns • Laying out planting stations • Digging out planting holes • Planting fruit trees • Growing and managing one crop from the following groups: Group A: Deciduous fruits, apple and peaches Group B: subtropical fruits: bananas, guavas, mangoes	• ICT tools with JAWS software • Fruit trees • Seedlings • Planting boards • Pegs • Lines • Digging tools • Organic matter • Inorganic fertilizers

	• Prepare fire guards • Identify signs of maturity • Market orchard fruits		Group C: Citrus fruits, oranges, naartjies • Preparing fire guards • Identifying signs of maturity • Marketing orchard crops	
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SUB TOPIC: CROP PROTECTION

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Pests	• describe one pest with a complete metamorphosis • describe one pest with an incomplete metamorphosis	• Life cycle of pests - Complete and incomplete metamorphosis	• Discussing the life cycle of a pest with a complete and incomplete metamorphosis • Drawing well labelled diagram of life cycle of a pest with complete and incomplete metamorphosis • Collecting and preserving pests' specimens	• ICT tools • Pest samples and specimens • Pictures of pests
Diseases	• discuss how plant diseases are spread • identify symptoms of named groups of diseases	• Transmission • General symptoms of: - Fungal - Bacteria	• Discussing the spread of plant diseases • Observing symptoms of plant diseases	• ICT tools • Diseased plants and specimens

		- Viral diseases	• Watching video clips and photos of various plant diseases	• Pictures of diseased plants
Weeds	• classify weeds as narrow and broad-leaved weeds • identify the mode of spread of common weeds	• Weeds • Mode of spread	• Classifying weeds • Identifying the modes of spread of common weeds	• ICT tools • Weeds samples and specimens • Weeds handbook

TOPIC 4: ANIMAL HUSBANDRY

SUB TOPIC: TYPES OF LIVESTOCK

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Ruminants and non-ruminants	• identify ruminant and non-ruminant animals • distinguish characteristics between ruminants and non-ruminants	• Ruminants: - Cattle, sheep and goats • Non-ruminants: - Horses, donkeys, pigs, rabbits, poultry	• Discussing characteristics of ruminants and non-ruminants • Observing animals to distinguish ruminants from non-ruminants	• ICT tools • Livestock

SUB TOPIC: ANATOMY AND PHYSIOLOGY

TOPIC	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Reproduction in poultry	• draw and label reproductive system of a hen and cock	• Reproductive parts • Egg	• Observing the male and female organs from slaughtered hen or cock	• ICT tools • Eggs • Samples and specimens of

	- describe the process of egg formation - state functions of each part of an egg - draw and label parts of an egg		- Drawing and labelling reproductive parts of a hen and cock - Tabulating the reproductive parts and their functions - Observing and identifying parts of a boiled egg	chicken reproductive systems
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SUB TOPIC: NUTRITION

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Livestock nutrients	- name the main nutrients required by farm livestock - explain functions of each nutrient - describe deficiency symptoms of each nutrient - identify sources of main nutrients	Livestock nutrients	- Carrying out simple tests for nutrients in available feedstuffs - Collecting and identifying samples of foodstuffs rich in carbohydrates and proteins - identifying symptoms of malnutrition in livestock	- ICT tools - Feedstuffs - Animals with nutrient deficiency symptoms

SUB TOPIC: SMALL LIVESTOCK PRODUCTION

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Broiler management	- select type of feeds from day old to slaughter - rear broilers - assess growth rates of broiler chickens - keep physical and financial records	- Broiler feeds - Management practices	- Preparing rations for broiler chickens of different ages - Brooding and rearing broilers - Weighing broilers regularly and illustrate graphically growth rate - Compiling physical and financial records	- ICT tools - Scale/balance - Broiler Feeds - Broilers
Slaughtering, processing and marketing	- demonstrate the slaughtering and dressing of broilers - calculate dressing out percentage - identify market for broilers	- Slaughtering - Dressing - Marketing	- Slaughtering and dressing broilers using different methods - Calculating dressing out percentage - Carrying out market research for broilers - Calculating profit and loss for a broiler enterprise	- ICT tools - Broilers

SUB TOPIC: ANIMAL HEALTH

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
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		(knowledge, skills, values and attitudes)		
Livestock diseases and hygiene	- explain causes of diseases - explain the modes of livestock disease transmission - justify the importance of hygiene - discuss methods of prevention and control of diseases	- Pathogens and other causes of diseases - Transmission of diseases - Sanitation	- Surveying on causes and transmission of diseases in animals - Discussing the importance of hygiene - Investigating and discussing remedies to prevent and control diseases - Cleaning and disinfecting poultry houses	- ICT tools - Vet officer

TOPIC 5: FARM STRUCTURES AND MACHINERY

SUB TOPIC: FARM IMPLEMENTS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Adjustments of animal drawn implements	carry out adjustments on animal drawn implements	- Adjustments: - Depth - Width	- Adjusting the depths and width of the mould board plough, cultivator - Adjusting depth of the harrow	- ICT tools - Mould board plough - Cultivator - Harrow

SUB TOPIC: FENCING

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Fencing materials and tools	- identify fencing materials and tools - discuss advantages and disadvantages of different fencing materials	Fencing materials and tools	- Selecting suitable fencing materials - Using tools safely and correctly	- ICT tools - Pictures - Actual tools - Fencing tools

SUB TOPIC: FARM ROADS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Siting of farm roads	- discuss factors to be considered when siting a farm road - list equipment needed when siting farm roads - describe characteristics of well sited farm roads	- Siting a farm road - Siting equipment	- Discussing factors considered in siting farm roads - Describing equipment needed for siting a farm road - Siting a farm road - Maintaining roads	- ICT tools - Resource person - Farm roads - Siting equipment

SUB TOPIC: HARNESSING

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Harnesses	- Differentiate breast band from collar harnesses - Identify and label parts of breast band and collar harnesses - describe materials used for harnesses - harness specific animals	- breast band and collar harness - Materials for making harnesses - Harnessing animals	- Drawing and labelling breast band and collar harnesses - Making harnesses using locally available materials - Harnessing specific animals	- ICT tools - Breast band and collar harnesses

TOPIC 6: AGRI-BUSINESS

SUB TOPIC: FARM RECORDS AND ACCOUNTS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Profit and loss Account	- explain the functions of profit and loss account - design a profit and loss account	Profit and loss or income and expenditure accounts	- Discussing role of a profit and loss account in Agriculture - Preparing a profit and loss account at a	- ICT tools - Farm records - Pictures

	- identify the income and expenditure of a farm enterprise - calculate the income from an agriculture enterprise		school agricultural enterprise Calculating farm income and expenditure of farm enterprises	
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SUB TOPIC: MARKETING

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Types of markets	- distinguish between controlled and uncontrolled markets - describe formal and informal markets for major crops and livestock in Zimbabwe	- Controlled and uncontrolled markets - Formal and informal markets	- Surveying on marketing of Agricultural products - Discussing types of markets for local agricultural products	- ICT tools - Local agricultural markets

SUB TOPIC: AGRICULTURAL COOPERATIVES

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Cooperatives	- explain benefits of agricultural cooperatives - identify problems associated with agricultural cooperatives	Agricultural cooperatives	- Discussing the benefits of agricultural cooperatives - Researching on the problems linked to local agricultural cooperatives - Dramatising problems of cooperatives	- ICT tools - Co-operatives officer

FORM 3

TOPIC 1: GENERAL AGRICULTURE

SUB TOPIC: LAND USE

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Physical planning farm	discuss the importance of physical planning farm	Physical planning farm	- Discussing the importance of physical planning farm - Sketching farm layouts with suggested farming activities	- ICT tools - Map templates
Crop rotation	- outline principles of crop rotation - design a four-crop rotation cycle - discuss the advantages of crop rotation	Principles of crop rotation	- Discussing the principles of a four-crop rotation - Designing a four -crop rotation cycle - Practising crop rotation in the school garden	ICT tools

SUB TOPIC: ENVIRONMENTAL FACTORS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Rainfall	- describe distribution, effectiveness, reliability and intensity of rainfall in Zimbabwe - explain the effects of distribution and intensity of rainfall on agricultural activities	- Distribution - Effectiveness - Reliability - Intensity of rainfall - Agriculture activities in relation to distribution and intensity of rainfall	Discussing how agricultural activities in Zimbabwe are influenced by rainfall	- ICT tools - Map of Zimbabwe showing rainfall distribution

SUB TOPIC: FORESTRY

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Timber harvesting and marketing	- describe methods of harvesting trees - discuss methods of treating timber - identify possible markets	- Harvesting - Treating - Marketing	- Describing methods of harvesting trees - Harvesting trees - Treating timber - Marketing timber - Compiling production records	- Creosote - Carbolinium - timber
Deforestation	describe causes of deforestation	- Deforestation - Afforestation	Discussing the effects of deforestation	ICT tools

	• explain effects of deforestation • suggest possible solutions to deforestation • explain importance of afforestation and reforestation	• Reforestation	• Surveying on the extent of deforestation within their locality • Discussing effects of deforestation • Identifying possible solutions to deforestation in their locality • Planting trees	• Pictures
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SUB TOPIC: WILDLIFE

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Indigenous knowledge systems in management of natural resources	• explain how cultural values and beliefs affect the management of natural resources • explain how principles of conservation and preservation of wildlife affect trading • describe effects of poaching • discuss biodiversity in relation to genetics, species and ecosystem diversity	• Conservation and preservation • Poaching • Biodiversity • Genetic, species and ecosystem diversity • Ecology	• Discussing how cultural values and beliefs affect the management of natural resources • Conducting class/school census based on totems that relate to animals found in the community • Visiting protected areas • Debating on wildlife trade at local and international levels	• Protected areas • Resource person • Textbooks • ICT tools

	describe habitats of wild animals		- Establishing a nature reserve to encourage biodiversity - Field studying of a habitat to determine animal and plant species composition	
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TOPIC 2: SOIL AND WATER

SUB TOPIC: SOIL FORMATION

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Weathering	discuss various forms of weathering	Forms of weathering	- Discussing the various form of weathering - Observing weathering - Demonstrating weathering	- ICT tools - Rocks

SUB TOPIC: SOIL TEXTURE, STRUCTURE AND PROFILE

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Soil textural classes	- List the eight textural classes - Determine textural class of a soil using a soil textural triangle	Eight textural classes in Zimbabwe	- Discussing the eight textural classes - Carrying an experiment on soil texture using the	- Sieves of varying sizes - ICT tools

			sieve method to determine soil classes	
Soil structure	- explain factors affecting soil structure - explain the importance of soil structure - identify methods of improving and maintaining good structure	Soil structure	- Explaining the importance of soil structure - Discussing factors affecting soil structure - Describing methods of improving and maintaining good structure - Carrying out field observations of different soil structures	- ICT tools - Soil samples

SUB TOPIC: SOIL TYPES

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Improvement of physical characteristics of soils	describe methods of improving different soil types	- Soil improvement: - Sand and clay soils	- Describing methods of improving different soil types - Manuring soils - Adding anthill soil to sand soil - Liming	- Lime - Organic matter

SUB TOPIC: SOIL CONSTITUENTS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Importance of soil components	- describe the composition of an agriculturally viable soil - explain the importance of each soil component - describe the types of soil water - explain movement of water in the soil - explain field capacity - explain the role of living organisms in the soil	- Soil components - Types of soil water - Movement of water - Field capacity - Soil macro and microorganisms	- Discussing the importance of soil components - Experimenting on the percentage composition of air, water, organic and inorganic matter in the soil - Collecting soil organisms and identifying them	- ICT tools - Soil samples

SUB TOPIC: SOIL TEMPERATURE

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Influence of soil temperature on plant growth and soil organisms	explain the effects of soil temperature on plant growth and soil organisms	Soil temperature	Discussing the effects of soil temperature on plant growth and soil organisms	- ICT tools - Soil samples

	- explain effects of extreme temperatures on various stages of crop growth - outline measures that can be taken to reduce the effects of extreme soil temperatures		- Experimenting on the effects of temperature on seed germination - Conducting field experiments on mulching, shedding and watering	
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SUB TOPIC: SOIL FERTILITY

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Fertiliser application	- describe different methods of fertiliser application - calculate fertiliser application quantities per given area - differentiate basal from top dressing	- Application methods - Calculations - Time of application	- Describing different methods of fertiliser application - Calculating fertiliser quantities - Applying organic and inorganic fertilisers	- Organic and inorganic fertilisers - Agronomist
Soil sampling	- explain the importance of soil sampling - Explain the principles of soil sampling - sample soil using at least one method	- Principles - Methods of soil sampling	- Discussing the importance of soil sampling - Explaining the principles of soil sampling - Sampling soils	- ICT tools - Soil sampling equipment

Soil pH and liming	- describe how soils are tested for pH - discuss the influence of soil pH - describe methods of correcting soil pH - identify types of lime - explain the importance of liming materials - explain the difference between lime and fertiliser - interpret the significance of pH values	- Soil pH - Liming	- Discussing the influence of soil pH on crop production - Determining soil pH values using pH meter and universal indicator - Demonstrating lime application	- ICT tools - pH meter - Universal indicator - Soil samples
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SUB TOPIC: SOIL EROSION AND CONSERVATION

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Conservation methods and structures	- describe methods of soil conservation on arable and grazing lands - describe the construction of basic conservation structures to standard dimensions	- Conservation methods - Conservation structures	- Describing methods of soil conservation on arable and grazing lands - Constructing and maintaining structures - Measuring dimensions of mechanical	- ICT tools - Pictures

			conservation structures - Reclaiming eroded areas in and around the school - Practising biological conservation	
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SUB TOPIC: WATER LOSS AND SOIL DRAINAGE

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Drainage and water logging	- describe drainage and water logging - explain the effects of waterlogged soils on crop growth - describe methods of improving drainage	Drainage and water logging	- Describing drainage and water logging - Explaining the effects of waterlogged soils on crop growth - Discussing methods of improving drainage - Field touring to identify signs of run off - Identifying drainage structures on the land - Constructing and maintaining drainage structures	- ICT tools - Soil samples
Leaching	explain the causes of leaching in arable lands	Leaching	Discussing the causes of leaching in arable lands	- ICT tools - Soil samples

	- explain the effects of drainage on loss of plant nutrients - describe methods of controlling leaching in arable lands		Experimenting on leaching levels of different soils	
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SUB TOPIC: WATER CONSERVATION

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Rain harvesting and storage	- describe methods of harvesting and storing water - describe different ground water sources	- Rain harvesting and storage - Ground water sources	- Harvesting rain water using various methods - Maintaining water harvesting structures - Identifying ground water sources	- ICT tools - Water reservoirs

SUB TOPIC: IRRIGATION

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Methods and types of irrigation	- describe methods of irrigation - explain different types of irrigation - explain the advantages and	- Methods of irrigation - Types of irrigation	- Describing methods and types of irrigation - Applying water to crops using at least one method of irrigation	- ICT tools - Models - Irrigation schemes

	disadvantages of each method of irrigation		Visiting irrigation schemes in their locality	
Choice of an irrigation system	discuss the factors affecting choice of an irrigation system	Factors affecting choice of an irrigation system	- Demonstrating different methods of irrigation - Designing an irrigation system	ICT tools

TOPIC 3: CROP HUSBANDRY

SUB TOPIC: STRUCTURE OF FLOWERING PLANTS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Plant anatomy and physiology	- identify parts of the internal structure of a root, stem and leaf - explain functions of tissues in a root, stem and leaf	- Tissue distribution in a: - root - stem - leaf	- Identifying the internal structure of a root, stem and leaf on a microscope - Drawing cross sectional diagrams of stem, root and leaf - Discussing the functions of root, stem and leaf	- ICT tools - Microscopes - Pictures - Root, stem and leaf samples

SUB TOPIC: PLANT PROCESSES

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Water and nutrient uptake	- describe the absorption of water by plants through osmosis - explain nutrient absorption by roots through active uptake - describe absorption of water by seeds	- Osmosis - Diffusion - Active uptake - Imbibition	- Discussing processes of osmosis, diffusion, active uptake and imbibition - Experimenting osmosis - Demonstrating imbibition by means of experiments	- ICT tools - Experimental apparatus - Pictures - Seed - potato samples
Transpiration	- discuss the role of transpiration stream - describe the role of stomata - explain how wilting occurs - outline factors affecting rate of transpiration	- Transpiration - Transpiration streams - Role of stomata in transpiration - Importance of transpiration - Wilting	- Discussing factors affecting rate of transpiration - Carrying out field experiments to demonstrate transpiration - Identifying plants that are under water stress	- ICT tools - Pictures - Experiment apparatus
Photosynthesis	- describe how plants photosynthesise - state the chemical and word equation for photosynthesis	- Gaseous exchange - Roles of chlorophyll, carbon dioxide and light	- Discussing the process of photosynthesis - Conducting experiments to	- ICT tools - Experiment apparatus

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
		Word and chemical equation for photosynthesis	demonstrate the need for carbon dioxide, light and chlorophyll	
Translocation and food storage	- explain the process of translocation - identify plant food storage organs - state the nature of food stored by plants	- Translocation - Food storage organs	- Describing the process of translocation - Demonstrating the process of translocation through the ring barking experiment - Selecting plants available in the locality and identify where they store plant food - Identifying plant storage organs	- ICT tools - Plant storage organs

SUB TOPIC: CROP IMPROVEMENT

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Crop breeding (Maize)	- state the importance of crop breeding - explain heterosis	- Crop breeding - Hybrids	Discussing the importance of crop breeding	- ICT tools - Plant breeder

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
	- differentiate between open pollination and controlled pollination - state the three types of crop hybrids - describe the production of single, double- and three-way hybrids	- Heterosis in plant breeding - Open and controlled breeding - Single, double- and three-way hybrids	- Identifying crop hybrids grown in the locality - Visiting a plant breeding station and observe how hybrids are produced - Demonstrating maize breeding	

SUB TOPIC: CROP PRODUCTION

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Land preparation	- state reasons for land preparation - describe steps taken during land preparation - describe primary and secondary tillage methods - describe conservation or minimum tillage techniques - identify implements and tools used	- Reasons for land preparation - Land preparation procedures - Tillage machinery, implements and tools	- Discussing reasons for land preparation - Carrying out tillage operations - Describing the tillage practices - Identifying implements and tools used - Touring farming areas	- ICT tools - Tillage implements and tools

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Legume and cereal production	- grow one cereal and one legume crop - carry out management practices on selected crops - describe harvesting of the crop - store harvested crops - market crops	- Suitable cultivars from: - Cereals: maize, sorghum, wheat - Legumes: groundnuts, field beans, soya beans - Seed rates - Management practices - Storage structures - Marketing	- Selecting crop cultivars - Carrying out crop management practices on selected crops - Identifying signs of crop maturity - Storing harvested crops - Marketing crops	- ICT tools - AGRITEX Officer

SUB TOPIC: CROP PROTECTION

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Pests	- explain methods of pest control - describe the advantages and disadvantages of each method	- Pest control methods - Pesticides	- Discussing methods of pest control - Describing advantages and disadvantages of pest control methods	- ICT tools - Empty pesticides containers - Pesticides - Spraying equipment

	- describe the significance of IPM in pest control - describe mode of action of the main groups of pesticides		- Describing the significance of IPM in pest control - Describing the mode of action of the main groups of pesticides - Controlling pests	
Diseases	- describe methods of disease control - identify the correct chemicals used to control crop diseases	Disease control methods	- Describing methods of disease control - Scouting for crop diseases - Collecting samples of crops damaged by crop diseases	- ICT tools diseased crops - fungicides
Weeds	- describe methods of weed control - differentiate herbicides as selective and non-selective - differentiate herbicides on the basis of timing of application - compare effectiveness of different weed control methods - calibrate a knapsack sprayer	- Weed control - Herbicides	- Describing methods of weed control - Discussing the basis of herbicides selectivity - Identifying samples of types of herbicides - Experimenting on the effectiveness of different methods of weed control - Calibrating a knapsack sprayer	- ICT tools - Empty herbicide containers - Herbicides - Spraying equipment

TOPIC 4: ANIMAL HUSBANDRY

SUB TOPIC: ANATOMY AND PHYSIOLOGY

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Digestive system of a ruminant and non-ruminant	- identify parts of the digestive system of a ruminant and non-ruminant - explain functions of parts of the digestive systems - distinguish the difference between digestive systems of a ruminant and non-ruminant	Anatomy and physiology of the digestive systems	- Explaining functions of parts of the digestive systems - Drawing and labelling digestive system of a named ruminant and non-ruminant - Discussing digestive systems - Examining the digestive systems from slaughtered animal	- ICT tools - Specimens of digestive systems

SUB TOPIC: NUTRITION

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Types of feeds	- classify feed stuffs - identify feed stuffs for each class of livestock	- Roughages - Concentrates - Straight feeds	- Classifying feedstuffs - Preparing balanced ration	- ICT tools - Samples of feeds

SUB TOPIC: SMALL LIVESTOCK PRODUCTION

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, and values attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Rearing of either rabbits or layers or indigenous chickens	• identify breeds of a named animal • choose a suitable housing site for a named animal • design and construct housing for a named animal • describe nutritional requirements of a named animal • manage the young one of a named animal	• Breeds • Housing • Nutritional requirements • Management practices	• Comparing characteristics of different breeds • Choosing an appropriate breed • Designing a plan of an animal house • Designing a feeding programme for a named animal • Brooding and rearing animals • Carrying out necessary management practices	• ICT tools • Small livestock • Feeds samples • Housing plans

SUB TOPIC: NON-RUMINANTS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Rearing of Non-ruminants: Pigs or donkeys	- identify the breeds of a named animal - describe housing systems - manage the named animal to maturity	- Animal breeds - Housing systems - Management practices	- Discussing management practices of a named animal - Rearing a named animal - Researching on breeds of pigs or donkeys	- ICT tools - Resource person - Pigs/donkeys

SUB TOPIC: ANIMAL HEALTH

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Notifiable livestock diseases	- recognise signs and symptoms of notifiable diseases - describe the control methods for notifiable diseases	Notifiable diseases	- Discussing one disease from the following groups: a) Bacterial diseases: anthrax b) Viral diseases: foot and mouth or new castle	- ICT tools - Vet Officer

			c) Protozoan diseases: trypanosomiasis, Discussing and administering disease prevention and control measures	
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SUB TOPIC: ANIMAL IMPROVEMENT GENETICS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Genetics	- explain the stages in mitosis and meiosis - outline the effects of the environment on genes	Mitosis and meiosis	- Describing mitosis and meiosis - Discussing the effects of the environment on genes - Observing slides on stages of mitosis and meiosis	- ICT tools - Microscope

TOPIC 5: FARM STRUCTURES AND MACHINERY

SUB TOPIC: FARM IMPLEMENTS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Maintenance	describe routine maintenance of farm implements	- Routine maintenance -plough -cultivator -harrow	- Describing routine maintenance of farm implements - Using and maintaining farm implements	- ICT tools - Farm implements or pictures

SUB TOPIC: FENCING

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Treatment of fencing materials	- describe the treatment of wooden fencing materials - describe treatment of metal fencing materials - treat fencing materials	Fencing materials treatment	- describing chemicals used in treating wooden and metallic fencing material. - discussing treatment of fencing materials - treating fencing materials	Treating chemicals
Fencing specifications	- Describe the standard specifications of fencing on farms - Fence specifications to	Fencing specifications -Spacing of poles -Spacing of strands	- Discussing fencing specifications to - Fencing specifications	Fencing poles and strands

SUB TOPIC: FARM BUILDINGS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Farm buildings	- identify materials used for the construction of farm buildings - describe the properties of building materials	- Farm building materials - Properties of building materials such as: - Quality - Durability - Strength - Fire resistance - Termite resistance - Temperature resistance	- Discussing building materials - Touring farm buildings	- ICT tools - Pictures of farm buildings - Farm buildings

SUB TOPIC: FARM ROADS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Features of farm roads	- describe features of a farm road - state the dimensions of different features on a farm road - repair a farm road	- Farm road features and dimensions - Repairs	- Discussing features of a farm road - Repairing farm roads - Touring farm roads	- ICT tools - Pictures of farm roads - Road repair implements

SUB TOPIC: APPROPRIATE TECHNOLOGY

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Irrigation pumps	- identify parts of a hand or power operated irrigation pump - discuss the working principles of an irrigation pump - describe the routine maintenance of a pump	- Parts of hand or power operated pumps - Pump working principles - Maintenance	- Identifying pump parts - Discussing working principles of a pump - Maintaining pumps	- ICT tools - Pumps - Pictures of pumps

TOPIC 6: AGRI-BUSINESS

SUB TOPIC: PRINCIPLES OF ECONOMICS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Opportunity cost and choices	explain the opportunities available for the farmer in agriculture industry	- Opportunities available to the farmer - Farmers choices - Opportunity cost	Identifying agriculture opportunities available for enterprising at the school	ICT tools

	- describe the factors influencing farmers' choices in agricultural enterprises - explain the concept of opportunity cost		- Choosing appropriate agricultural enterprises for the school - Simulating the concept of opportunity cost - Identifying opportunity costs in real life situations	
Demand, supply and price	- describe the laws of demand and supply - interpret demand and supply curves/schedules - describe determinants of market price for agricultural commodities	- Demand and supply - Market price	- Describing the laws of demand and supply - Interpreting demand and supply curves and schedules - Demonstrating effects of price change on demand and supply - Surveying on demand and supply levels of farm produce at a local market	- ICT tools - Money - Local markets

SUB TOPIC: FARM BUDGETING

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Budgets	- explain the importance of budgeting in farming - identify sources of information for budgeting - calculate the gross margin for an agricultural enterprise - prepare partial and complete budgets	Budgeting	- Discussing the role of budgets in farming - Surveying on farm budgets - Drawing up gross margin, partial and whole farm budgets from sourced information	- ICT tools - Money - Source documents

SUB TOPIC: AGRICULTURAL MARKETING

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Functions and factors of marketing	- explain the functions of marketing in agriculture - state factors affecting marketing of agricultural produce	- Marketing functions - Factors affecting agricultural marketing	- Preparing agriculture produce for marketing - Selling agricultural produce to the market - Preparing agriculture produce for marketing - Selling agricultural produce to the market	- ICT tools - Agriculture produce - Local markets

			• Discussing functions of marketing in agriculture • Identifying factors affecting agricultural marketing	
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FORM 4

TOPIC 1: GENERAL AGRICULTURE

SUB TOPIC: ENVIRONMENTAL FACTORS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Natural disasters	explain the effects of natural disasters on Agriculture	- Hailstorm - Floods - Cyclones - Whirl wind - Veld fire - Drought - Heat wave	- Discussing effects of natural disasters on agriculture - Watching videos of natural disasters - Visiting Met Stations	- ICT tools - Pictures - Weather station
Disaster risk management strategies	outline precautionary measures to guard against natural disasters	- Weather forecast - Disaster preparedness - Conservation structures	- Discussing the measures that can be used to guard against natural disasters - Constructing conservation structures around their community - Listening to weather reports	- ICT tools - Pictures

SUB TOPIC: FORESTRY

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Agro-forestry	- discuss the importance of agro-forestry practices - identify agro-forestry components - establish agro-forestry plots	Agro-forestry practices	- Discussing the importance of agroforestry practices - Identifying agro-forestry components - Designing and implementing an agro-forestry project at their school	- ICT tools - Pictures - Forestry officer

SUB TOPIC: WILDLIFE

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Human and wildlife conflicts	- discuss possible conflicts between humans and wildlife - explain the existing legislation in managing wildlife resources in Zimbabwe - discuss the role of Government and voluntary	- Human and wildlife conflicts - Protection of resources - Legislation - Government and voluntary organisations - International conventions	- Discussing possible conflicts between humans and wildlife - Role playing depicting conflicts between humans and wildlife - Collecting relevant information on Government Policy as regards to wildlife management	- ICT tools - Pictures - National Warden Parks

	organisations in wildlife management • discuss the role of international conventions in wildlife management		• Debating the role of Government and voluntary organisations in wildlife management	
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TOPIC 2: SOIL AND WATER

SUB TOPIC: SOIL FERTILITY

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Nitrogen cycle	• describe the nitrogen cycle with the aid of a diagram	• Nitrogen cycle	• Describing the nitrogen cycle • Illustrating the nitrogen cycle	• ICT tools

SUB TOPIC: WATER CONSERVATION

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Water pollution	• explain causes of water pollution • discuss ways of reducing water pollution	• Water pollution	• Discussing causes of water pollution • Collecting water samples to determine levels of pollution	• Textbooks • ICT tools

	describe the effects of water pollution on agricultural production		Discussing ways of reducing water pollution	
Water legislation	- discuss water use legislation - discuss water management in Zimbabwe	Management of national water (ZINWA)	Identification of water bodies	ICT tools

SUB TOPIC: IRRIGATION

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Irrigation equipment	- identify irrigation equipment - describe the structures and functions of irrigation equipment	- Irrigation equipment: - Structures and functions	- Assembling of irrigation equipment - Discussing structures and functions of irrigation equipment	- ICT tools - Pictures of irrigation equipment - Irrigation equipment

TOPIC 3: CROP HUSBANDRY

SUB TOPIC: PLANT PROCESSES

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Respiration	- state the word and chemical equation for aerobic respiration - distinguish between aerobic and anaerobic respiration - identify sites of respiration - describe the importance of respiration - explain the differences between photosynthesis and respiration	Respiration	- Illustrating aerobic respiration using word equation - Experimenting on respiration - Comparing aerobic and anaerobic respiration - Identifying sites of respiration - Describing importance of respiration - Comparing respiration and photosynthesis	- ICT tools - Experimental apparatus
Plant tropisms	- describe each of the tropisms - demonstrate the responses of plant parts to tropisms - evaluate the importance of plant tropisms	- Plant tropisms - Responses to light, touch, gravity and water	- Describing the plant tropisms - Demonstrating each tropism through experiments - Discussing the importance of plant responses	- ICT tools - Experimental apparatus

SUB TOPIC: CROP PROTECTION

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Agrochemicals	- explain the precautionary measures taken when using and storing chemicals - describe the toxicity levels of agro-chemicals - calculate the mixing ratios of chemicals used in spraying - draw and label parts of a knapsack sprayer	- Agro-chemicals - Toxicity levels - Knapsack sprayer	- Reading instructions on chemical labels - Identifying the formulation of pesticide - Determining the strengths of formulation - Identifying parts of a knapsack sprayer	- ICT tools - Agro-chemicals samples - Knapsack sprayers - Pictures showing agro chemicals

TOPIC 4: ANIMAL HUSBANDRY

SUB TOPIC: ANATOMY AND PHYSIOLOGY

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Reproductive system of a ruminant	- identify the parts of the reproductive systems of the male and female ruminant - explain functions of parts of the reproductive systems - draw and label the reproductive parts of male and female ruminant	Reproductive systems of male and female ruminants	- Drawing and labelling reproductive systems of a named male or female ruminant - Discussing functions of parts of the reproductive systems	- ICT tools - Specimens of male and female ruminant reproductive systems - Pictures of male and female ruminant reproductive systems

SUB TOPIC: NUTRITION

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Maintenance and production rations	calculate maintenance and production rations	Maintenance and production rations	Calculating production ration using Pearson Square method	- ICT tools - Animal nutritionist - Animal feeds

	prepare rations for both ruminants and non-ruminants		Mixing rations to produce a balanced diet	
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SUB TOPIC: SMALL LIVESTOCK PRODUCTION

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Slaughtering, processing and marketing	- slaughter and dress rabbits or off-layers or indigenous chickens - prepare pelts or eggs for market - compile financial and production records	Slaughtering and marketing of rabbits/layers/indigenous chickens	- Slaughtering and dressing rabbits or off-layers - Packaging and refrigerating rabbits or off-layers or indigenous chickens - Identifying suitable market - Compiling financial and production records	- ICT tools - Slaughtering materials - Small livestock

SUB TOPIC: RUMINANTS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Management of cattle or sheep or goats	list exotic and indigenous breeds in Zimbabwe	Exotic and indigenous breeds	Discussing characteristics of	- ICT tools - Pictures of breeds

	- describe characteristics of exotic and indigenous breeds - describe management practices	Management practices	- exotic and indigenous breeds - Discussing management practices - Researching on animal breeds - Conducting a tour of animal rearing farms	- Resource person - Vet officer - livestock
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SUB TOPIC: ANIMAL HEALTH

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Animal parasites and immunisation	- identify internal and external parasites of farm livestock - describe the life cycles of one internal and one external parasite - prevent and control parasites - explain the Animal Health Act - describe the types of livestock immunity	- Parasites: - Life cycle - Symptoms - Prevention - Treatment - Control - The Animal Health Act - Immunity	- Identifying external and internal parasites of livestock - Discussing the life cycle of one host tick and a roundworm - Discussing the prevention and control of internal and external parasites - Explaining the Animal Health Act - Discussing the types of immunity in livestock	- ICT tools - Pictures - Vet officer - Specimens of parasites - Animal Health Act

SUB TOPIC: ANIMAL IMPROVEMENT GENETICS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Breeding	- explain the importance of breeding - explain the effects of the environment on breeding - distinguish cross-breeding from in-breeding - select animals for breeding	- Livestock breeding - Effects of the environment on breeding - Types of breeding - Importance of artificial selection	- Discussing the importance of breeding - Discussing the effects of the environment on breeding - Drawing genetic diagrams in test cross problems - Selecting animals for breeding	- ICT tools - Animal breeder

TOPIC 5: FARM STRUCTURES AND MACHINERY

SUB TOPIC: FENCING

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Anchors and fencing calculations	- explain the role of anchors - outline the advantages and disadvantages of different anchors	- Anchors - Fencing calculations	- Explaining the role of anchors - Calculating quantities of materials required per given perimeter	- ICT tools - Anchors

	- determine the quantities of materials required per given perimeter - construct anchors		- Discussing the advantages and disadvantages of different anchors - Constructing anchors	
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SUB TOPIC: FARM BUILDINGS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Designing livestock buildings	- draw plans of buildings suitable for livestock - calculate costs of construction - determine the cost effectiveness of each material	- Livestock building designs - Costing	- Designing livestock houses - Calculating costs - Costing building materials	- ICT tools - Building plans

SUB TOPIC: FARM ROADS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Road construction and maintenance	identify materials required for the construction of a farm road	- Road construction - Farm road maintenance	Mobilising materials required for construction of a farm road	- ICT tools - Roads

	- describe the construction of a farm road - maintain local farm roads		- Discussing the construction of a farm road - Maintaining farm roads	
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SUB TOPIC: APPROPRIATE TECHNOLOGY

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Shellers	- identify parts of a maize and groundnut sheller - explain functions of each part - explain operational principles of shellers	- Parts of shellers - Operation principles	- Describing functions of maize /groundnut shellers parts - Discussing the operational principles of a sheller - Using shellers appropriately	- ICT tools - Shellers

TOPIC 6: AGRI-BUSINESS

SUB TOPIC: PRINCIPLES OF ECONOMICS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Diminishing returns	- explain the law of diminishing returns - interpret the law of diminishing returns - describe the implications of diminishing returns in agriculture	- Law of diminishing returns - Agriculture implications of diminishing returns	- Discussing the law of diminishing returns - Interpreting the law of diminishing returns from graphs - Watching video simulations on the effects of increased inputs on outputs while other factors are held constant	ICT tools
Risk and uncertainty	- Outline risks and uncertainties that can be encountered in Agriculture - distinguish between risks and uncertainties - explain ways of minimising the effects of risks and uncertainties	- Differences between risks and uncertainties - Avoiding risks and uncertainties	- Discussing risks and uncertainties - Conducting a survey to assess risks and uncertainties on the school or community farms - Discussing ways of minimising the effects of risks and uncertainties identified	ICT tools
Decision making	explain the importance of	Importance of decision making	Discussing the importance of	ICT tools

	decision making in agriculture • outline the economic factors influencing decision making • outline the steps to follow when making decisions on a farm.	• Economic factors in decision making • Decision making process	decision making in agriculture enterprises • Identifying economic factors influencing decision making	
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SUB TOPIC: MARKETING

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Marketing legislation	• explain the marketing legislation for agricultural produce and commodities	• Changes in marketing legislation on agricultural products	• Identifying crop and animal products that are controlled by marketing legislation in Zimbabwe	• ICT tools • Newspapers • Magazines • Pictures • Flyers • Brochures

8.0 ASSESSMENT

Learners shall be assessed through School Based Continuous Assessment (SBCA) and Summative Assessment (SA). These assessments shall be guided by the principles of inclusivity, practicability, authenticity, transparency, flexibility, validity and reliability. The principles are crucial for creating a supportive and effective learning environment that fosters growth and development in learners. Arrangements, accommodations and modifications shall be visible to enable candidates with special needs to access assessments.

This section covers the assessment objectives, the assessment model, the scheme of assessment, and the specification grid.

8.1 ASSESSMENT OBJECTIVES

Learners will be assessed on their ability to:

- 8.1.1 outline control measures of pests, diseases and parasites of a named agricultural enterprise
- 8.1.2 interpret and evaluate the profitability of a given agricultural enterprise
- 8.1.3 describe the anatomy and physiology of named plants and animals
- 8.1.4 illustrate breeding processes in plants and animals
- 8.1.5 outline the effects of the environment on crop and animal production
- 8.1.6 discuss the production of a named plant and animal up to marketing
- 8.1.7 outline the socio-economic importance of agriculture to the family, community and the nation
- 8.1.8 discuss health and safety measures in agriculture
- 8.1.9 correctly use and maintain any given agricultural equipment
- 8.1.10 describe and maintain named agricultural structures
- 8.1.11 design experiments and correctly interpret the results
- 8.1.12 apply acquired skills, knowledge and information to solve agricultural problems in the community
- 8.1.13 maintain accurate physical and financial records of a named agricultural enterprise
- 8.1.14 explain the socio-economic importance of wildlife and forestry
- 8.1.15 justify the importance of land tenure on agricultural activities in Zimbabwe

8.2 Assessment Model

Assessment of learners shall be both Continuous and Summative as illustrated in Figure 1. School Based Continuous Assessment shall include recorded activities from the School Based Projects done by the learners. The mark shall be included on learners' end of term and year reports. Summative assessment at school level shall include terminal examinations which are at the end of the term and year.

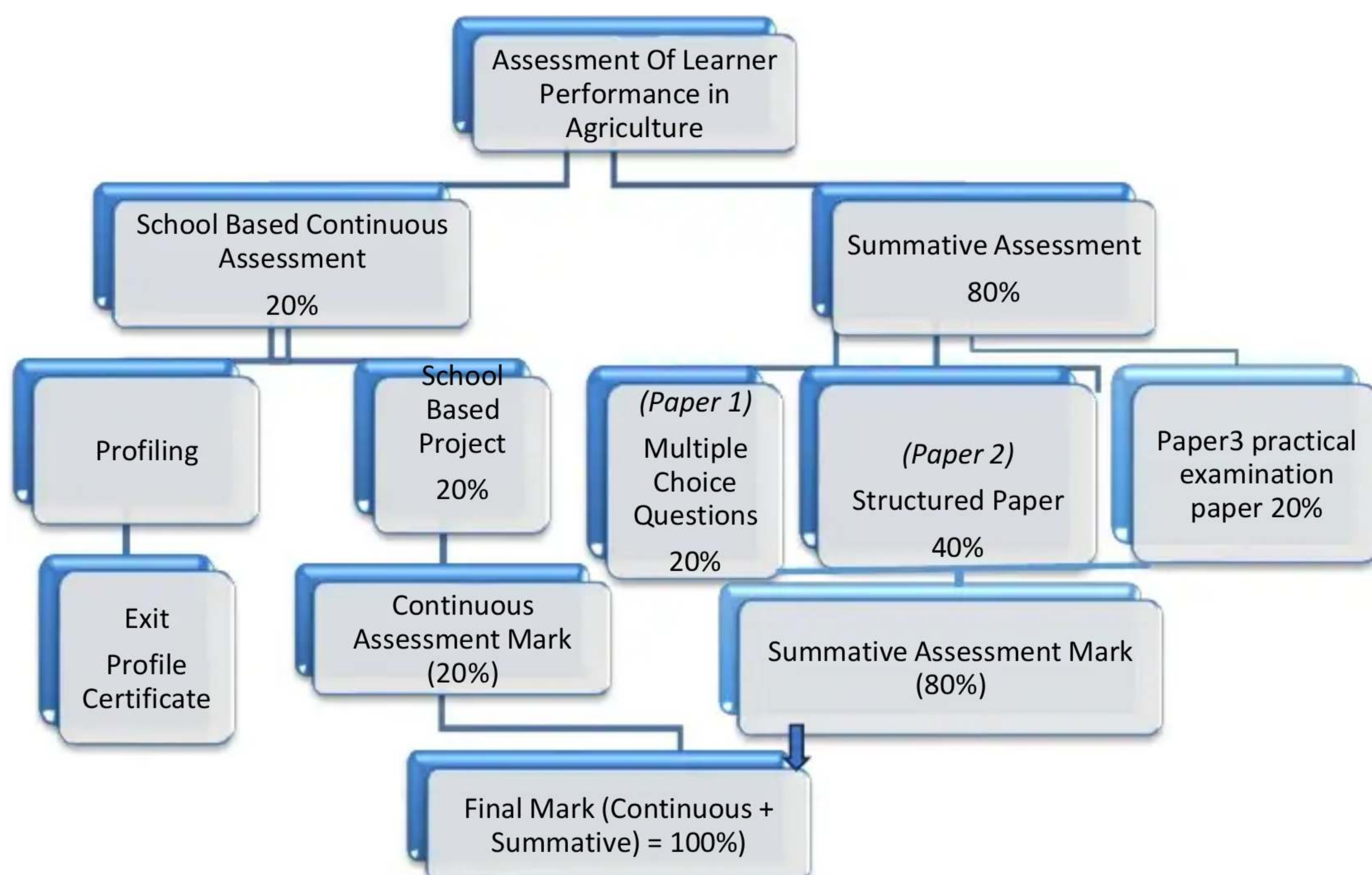


Fig. 1 Assessment Model

In addition, learners shall be profiled and learner profile records established. Learner profile certificates shall be issued for checkpoints assessment in schools as per the dictates of the Teacher's Guide to Learning and Assessment. The aspects to be profiled shall include learner's prior knowledge, values and skills, and subsequently the new competences acquired at any given point.

8.3 Scheme of Assessment

The Assessment Model shows that learners shall be assessed using both School Based Continuous Assessment and Summative Assessment for both School and

ZIMSEC assessments.

The table shows the Scheme of Assessment where 20% is allocated to School Based Continuous Assessment and 80% to School or ZIMSEC Summative Assessment.

FORM OF ASSESSMENT	WEIGHTING
School Based Continuous Assessment	20%
Summative Assessment	80%
Total	100%

8.3.1 Description of School Based Continuous Assessment

Learners shall do one school-based project per form which contributes to 20% of the end of year final mark. The end of year summative assessment shall then contribute 80%. However, for ZIMSEC public examinations, two (2) school-based projects shall be considered as School Based Continuous Assessment at Form 6. The two School Based Projects shall include those done during Form 3 and 4 sessions. Each will contribute 10%.

8.3.1.1: School – Based Project Continuous Assessment Scheme

The Table given below shows the Learning and Assessment Scheme for the School Based Project.

Project Execution Stages	Description	Timelines	Marks
1	Problem Identification	January	5
2	Investigation of related ideas to the problem/innovation	February	10
3	Generation of possible solutions	March	10
4	Selecting the most suitable solution	April-May	5
5	Refinement of selected solution	June	5
6	Presentation of the final solution	July	10
7	Evaluation of the solution and Recommendations	August-September	5
	TOTAL		50

8.3.2 Description of the ZIMSEC Summative Assessment

ZIMSEC Summative Assessment shall be a public examination at Form 4. The examination shall consist of three (3) papers.

Paper 1 (1hr – 40 marks) 20%

There are **40 Objective** type questions and candidates are required to answer all.

Paper 2 (2hrs – 100 marks) 40%

Paper 2 consists of three sections.

Section A: Six compulsory structured questions based on the whole syllabus - 60 Marks.

Section B: Three essay type questions will be set on Crop Production and candidates must **answer one** question only. Each question carries (20) marks. Candidates are expected to show thorough understanding of practical skills involved in the studied areas.

Section C: Three essay type questions will be set on Animal Production and candidates must **answer one** question only. Each question carries (20) marks. Candidates are expected to show thorough understanding of practical skills involved in the studied areas.

Paper 3 Practical 20%

This is a practical paper set from any part of the syllabus. The paper will be based on experiments, investigations, observations and calculations. Full instructions will be given where unfamiliar material or techniques will be required. Two compulsory questions are set. Each question carries 20 marks.

Paper total 40 marks.

NOTE: A profile system has to be developed for every learner to capture those attributes that cannot be measured such as the soft skills. A folio comprises test results throughout the secondary school on an annual basis and marks collected from the four prescribed projects. Observation schedules, checklists, tests and project tasks are to be set at district level and standardised nationally.

8.4 Specification Grid

Skill	Paper 1	Paper 2	Paper 3
Knowledge and comprehension	45%	45%	30%
Application and Analysis	40%	40%	50%
Problem solving	15%	15%	20%
TOTAL	100%	100%	100%